

The Ultimate ML Learning Path

We are going to tackle these in 4 Phases. Do not jump ahead! Master one phase before moving to the next.

Phase 1: The Foundations (Supervised Starters)

These are your bread-and-butter algorithms. They are simple, interpretable, and used everywhere.

1. **Linear Regression** (Regression)
 - *The Goal:* Predicting specific numbers (e.g., "What will the house price be?").
 - *The Twist:* **Ridge & Lasso Regression**. Learn these right after. They are just Linear Regression with a penalty to stop the model from memorizing data too closely (overfitting).
2. **Logistic Regression** (Classification)
 - *The Goal:* Predicting Yes/No answers (e.g., "Is this customer churning?").
3. **Naïve Bayes** (Classification)
 - *The Goal:* Great for text and speed (e.g., "Is this email spam?").
4. **K-Nearest Neighbors (KNN)** (Classification & Regression)
 - *The Goal:* The lazy learner. It just looks at what similar data points are doing.

Phase 2: The Logic Builders (Non-Linear)

Now we move away from straight lines and start thinking like a human making decisions.

5. **Decision Trees** (Classification & Regression)
 - *The Goal:* A flowchart of rules.
 - *Note:* Learn how it works for both **Regressors** (predicting numbers) and **Classifiers** (predicting groups).
6. **Support Vector Machines (SVM)** (Classification)
 - *The Goal:* Drawing the widest possible street (margin) between two groups of data.

Phase 3: The Powerhouses (Ensemble Methods)

This is where you win competitions. We combine multiple weak models to make one strong one.

7. **Random Forest** (Ensemble)
 - *The Goal:* A voting system. You grow lots of Decision Trees and let them vote on the answer.
8. **Gradient Boosting (XGBoost)** (Ensemble)
 - *The Goal:* The smart fixer. It builds trees one by one, where each new tree tries to fix the mistakes of the previous one.
 - *Focus:* Keep this conceptual! You don't need heavy math here, just understand *how* it boosts performance.

Phase 4: The Explorers (Unsupervised)

Finding patterns when you don't have the answers (labels).

9. K-Means Clustering (Clustering)

- *The Goal:* Grouping similar things together (e.g., "These customers shop similarly").

10. Principal Component Analysis (PCA) (Dimensionality Reduction)

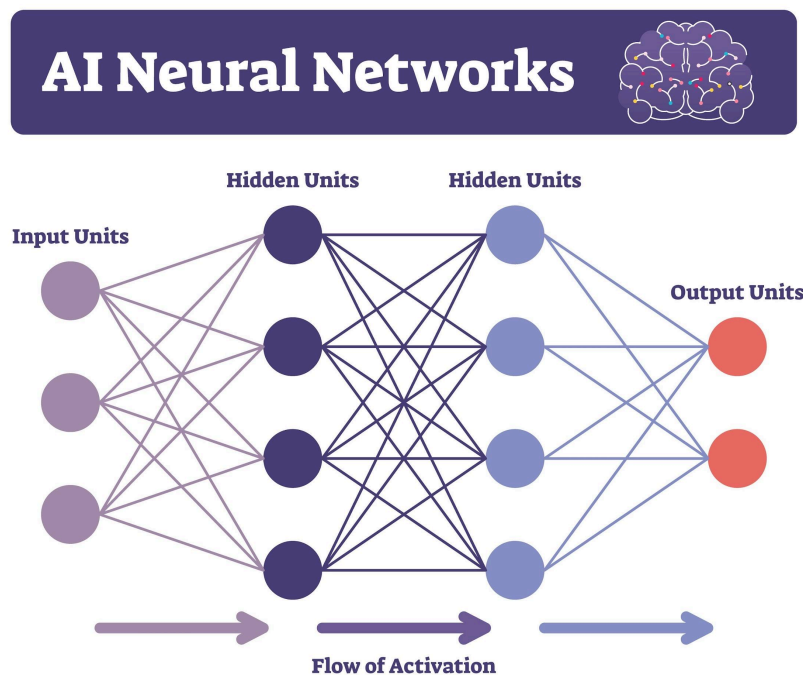
- *The Goal:* Simplifying complex data. It takes many messy columns and squashes them into a few important ones.

🛑 STOP HERE (Complete "Big 10" First) 🛑

The phases below are the "Advanced Expansion."

Phase 5: The Brain (Deep Learning)

This is where we leave standard statistics and enter "AI".



- **Neural Networks (ANNs):** The foundation of Deep Learning.
 - **CNNs:** For images (Computer Vision).
 - **RNNs:** For sequence data (Time series).
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Phase 6: The Modern Giant (GenAI & LLMs)

The technology behind ChatGPT, Gemini, and content generation.

- **Transformers:** The architecture that changed everything.
 - **LLMs:** Understanding how Large Language Models predict text.
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Phase 7: The Specialist (Reinforcement Learning)

Training agents through rewards. Very niche.

- **Q-Learning:** The basics of teaching a robot/agent through trial and error.
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What To Learn For Each Algorithm (VERY IMPORTANT)

Algorithm **TODO Template:** For **every algorithm**, follow the **same** checklist

1. Problem type (Regression / Classification)
 2. Intuition (How it works in simple words)
 3. Assumptions
 4. When to use
 5. When NOT to use
 6. Strengths
 7. Weaknesses
 8. Important hyperparameters
 9. Overfitting behavior
 10. One real-world example
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The "To-Do" Checklist

For **every single algorithm** above, you must be able to answer these 5 points in simple English. If you can't, you aren't ready to move on!

1. **The "Elevator Pitch" (Concept):**
 - Can you explain it to a 10-year-old?
 - *Example:* "KNN is like asking your neighbors for a movie recommendation. If most like action, you'll probably watch action."
 2. **The "Mathy" Bit:**
 - You don't need the Ph.D. derivation, just the core equation.
 - *Example:* For Linear Regression, know $y = mx + c$.
 3. **The Assumptions (The "Gotchas"):**
 - What does the model need to work?
 - *Example:* Linear Regression assumes the data follows a straight line.
 4. **Pros & Cons:**
 - Why use it? (e.g., "It's super fast.")
 - Why avoid it? (e.g., "It gets confused by outliers.")
 5. **Hyperparameters (The Knobs):**
 - What settings can you tweak to improve the model? (e.g., "K" in KNN).
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